Navbea RTLS Solutions Overview

Real-Time Location and IoT Innovations for Enhanced Safety and Efficiency

navbea



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Company Introduction

Experienced Workforce

Navbea boasts a skilled team certified in IoT-based efficiency and occupational health and safety applications, driving cutting-edge technological development.

International Projects

Successfully implemented RTLS systems in major theme parks and thermal power plants across Azerbaijan, with ongoing expansion into global markets.

Innovative Solutions

Pioneering AI computer vision and deep learning technologies integrated into occupational safety and real-time location tracking platforms.

European Office

Opened a strategic office in Amsterdam, Netherlands, to support international growth and enhance customer engagement across Europe.



Platform

What is Navbea RTLS?

Navbea RTLS is an advanced Internet of Things (IoT) and Real-Time Location System (RTLS) platform designed to monitor and manage personnel, equipment, and sensor data across various environments.

The platform integrates smart cards, sensors, and readers to collect real-time data, which is then processed and analyzed on the Navbea platform. This allows for live tracking, reporting, and automated auditing based on predefined rules to improve operational workflows, safety measures, and resource management.

By leveraging IoT connectivity and real-time location technology, Navbea delivers actionable insights that help organizations optimize workforce deployment, equipment usage, and environmental monitoring efficiently and effectively.

RTLS Technology Components



Smart Cards and Sensors

Smart cards and sensors capture real-time data on personnel and equipment location and status for continuous monitoring.



Readers and Transmitters

Readers receive signals from smart cards and sensors, transmitting data to the central monitoring system instantly and reliably.



Navbea Platform

The Navbea platform processes incoming data in real-time, providing live tracking, automated rule auditing, and detailed reporting.



Data Processing & Monitoring

Data is analyzed continuously to enable immediate alerts, operational insights, and decision-making support for facility management.

Solutions

Core RTLS Solutions

Asset & Inventory Management

Uses active and passive sensors to track equipment and inventory in real-time, ensuring accurate stock levels, quick location, and optimized asset utilization.

Personnel Positioning & Safety

Real-time tracking of personnel via wearable sensors for safety monitoring, emergency alerts, assembly area counts, and lone worker protection.

Vehicle & Forklift Management

GPS and AI-enabled management for vehicles and forklifts including task assignments, live location tracking, and efficiency optimization to reduce fuel consumption.

Visitor & Subcontractor Tracking

Monitors subcontractor and visitor movements with access control and alarm systems to enhance security and compliance within facilities.

Safety and Emergency Features

Emergency Button & Instant Alerts

Personnel can press an emergency button on their smart cards to send instant notifications via SMS, email, and platform alerts to relevant safety units, enabling swift response.

Assembly Area Monitoring & Quick Count

In emergencies, real-time tracking of personnel at assembly points allows quick roll calls and identification of missing individuals for rapid intervention.

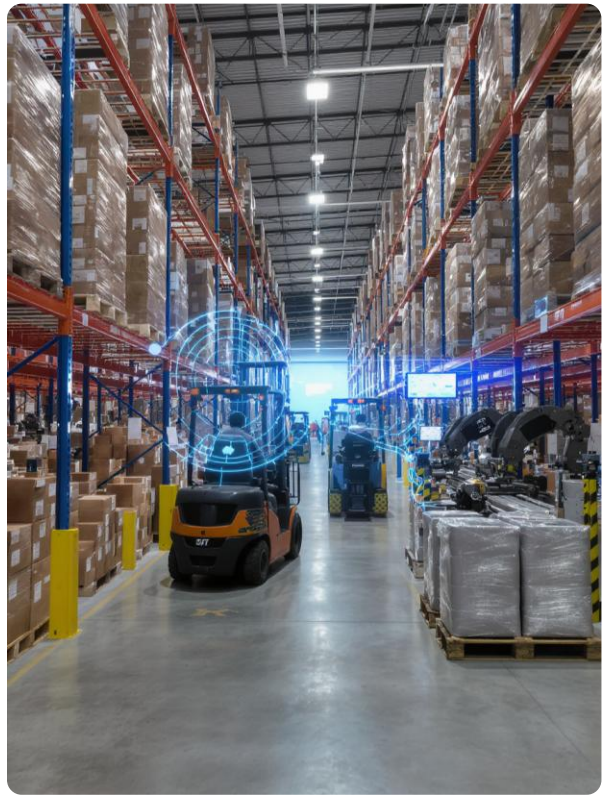
Inactivity Detection & Lone Worker Safety

Wearable sensors detect inactivity or falls, triggering early warnings to prevent accidents and ensure lone workers receive timely assistance.

Efficiency

Equipment Tracking and Efficiency

Real-time equipment tracking boosts efficiency by enabling quick location, usage analysis, maintenance alerts, and controlled access to reduce loss and downtime.



Equipment Tracking Benefits

Navbea's solution offers real-time equipment location and usage monitoring. It enables quick finding, usage analysis, and timely maintenance alerts to prevent failures. Access control restricts unauthorized use, reducing loss and downtime.

Key Features

- Real-time location tracking for quick equipment retrieval
- Usage analysis to optimize operations
- Maintenance alerts to prevent failures
- Access control to limit unauthorized use

Risk and Emergency Management

Navbea’s Risk and Emergency Management solutions provide critical support during emergencies by enabling real-time tracking of personnel and assets. When an emergency button on personnel cards is pressed, notifications are instantly sent via SMS, email, and platform alerts to relevant units. The system facilitates rapid search and rescue by identifying the last known locations of individuals and equipment. Additionally, real-time monitoring of personnel access to assembly areas ensures quick roll call and intervention, improving overall emergency response efficiency and safety. These features help organizations minimize risks and streamline emergency management processes.



Asset & Inventory Circularity Management

Active Sensors

Active sensors have their own power source and regularly send signals, enabling continuous real-time tracking of assets and products throughout their lifecycle.

Passive Sensors

Passive sensors are low-cost and draw power externally, used for efficient RFID and barcode scanning to support automatic stock counting and product location tracking.

Inventory Accuracy

Automated counting through sensors ensures high stock accuracy, reducing human error and enabling faster inventory control processes by up to 70%.

Circular Economy Support

Sensor-based tracking optimizes asset utilization and minimizes lost inventory, promoting sustainable resource use and supporting circular economy initiatives.

Operations

Business Process and Time Management

Navbea's business process and time management solution leverages micro-location technology to assign specific tasks to personnel and equipment based on their exact location within a facility. This enables precise monitoring of task progress, shift durations, and overall workforce performance through a centralized digital platform. The system collects real-time data on task completion times and shift tracking, allowing managers to identify inefficiencies and optimize resource allocation. By systematically monitoring operations, businesses can minimize downtime, reduce operational costs, and improve workforce planning, resulting in enhanced productivity and streamlined workflows.

Visitor and Subcontractor Tracking

Subcontractor and Visitor Monitoring

- Smart cards track subcontractor personnel and guest movements within designated areas.
- Monitoring of subcontractor working hours ensures compliance and auditing capabilities.
- Enhances security by controlling access to sensitive zones.
- Real-time data supports quick response to any unauthorized presence.

Unauthorized Access Alarm System

- Instant alerts trigger when unauthorized individuals attempt to enter restricted areas.
- Automatic notifications sent to relevant security units via SMS and web platforms.
- Alarm system integrates with the Navbea platform for centralized control.
- Strengthens overall facility security by preventing breaches proactively.

Environmental Monitoring and Cold Chain Tracking

Navbea’s Environmental Monitoring and Cold Chain Tracking solution centralizes the collection, monitoring, and reporting of critical environmental values using a variety of sensors. This system tracks temperature, humidity, and other parameters from a single platform to maintain ideal storage conditions and ensure product quality. When environmental values exceed predefined thresholds, the platform automatically generates alarms to alert relevant personnel, enabling immediate corrective action. This real-time monitoring capability supports compliance with regulatory standards and enhances operational efficiency by preventing spoilage and equipment failure in cold chain logistics and sensitive environments.

Occupational Health and Safety Solutions



AI Camera Monitoring

AI-enabled cameras installed on forklifts and in key areas detect pedestrians and vehicles, triggering automatic slowing and audible warnings to prevent collisions and accidents.

Wearable Wristband Alerts

Sensor-equipped wristbands vibrate to alert operators when entering hazardous zones, ensuring personal safety through proactive warning systems.

Pedestrian Detection & Accident Prevention

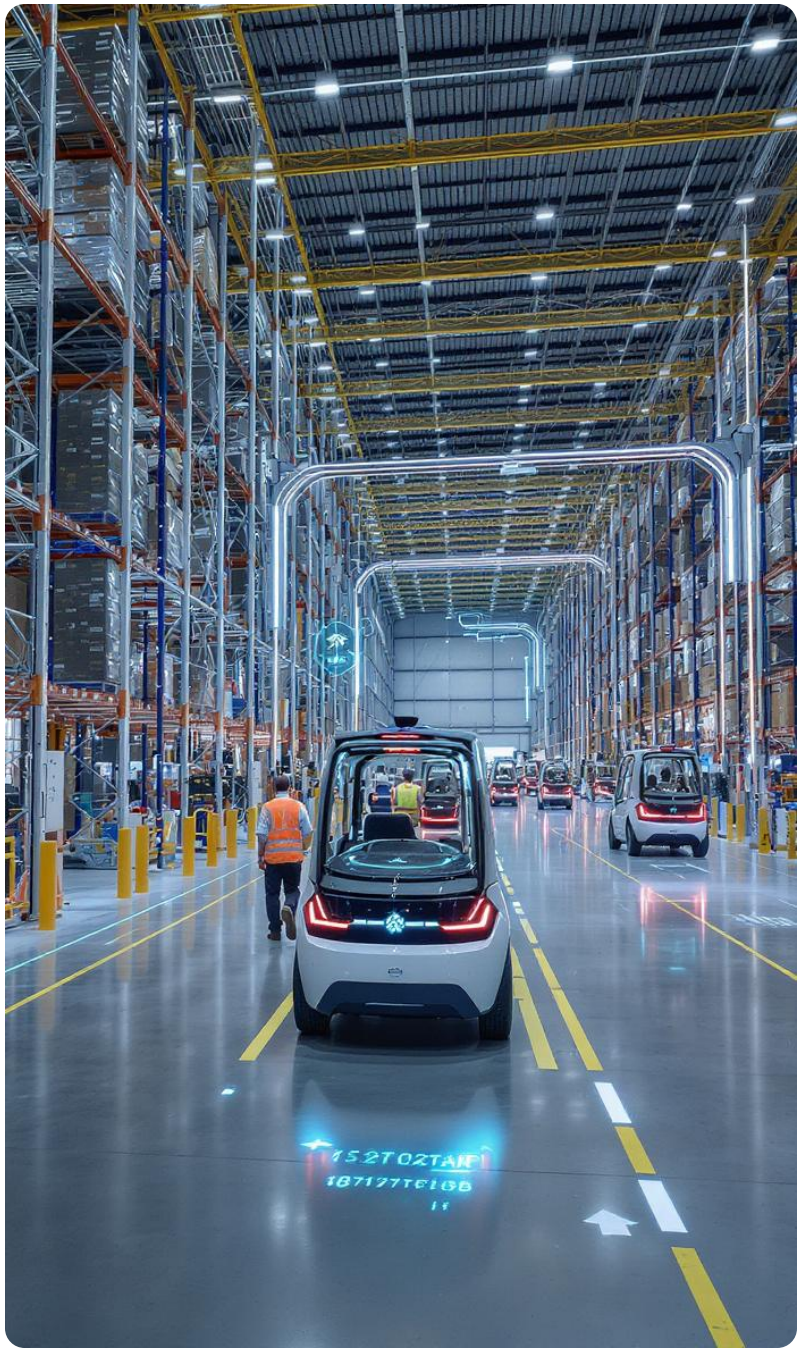
Multi-layered safety systems combine AI, sensors, and real-time data to minimize workplace accidents by monitoring movement and enforcing restricted area controls.

Efficiency

Vehicle, Forklift, and Ramp Management

Navbea’s Vehicle, Forklift, and Ramp Management solutions use GPS and AI to streamline operations. Real-time tracking monitors movements while AI automates task assignments based on location and priority. This reduces waiting times, cuts fuel use, and boosts efficiency.

Detailed reports help optimize routes and vehicle usage, saving costs and resources. Intelligent monitoring enhances safety by preventing collisions and ensuring protocol compliance, making logistics safer and more effective.



AI-Powered Monitoring Features

Advanced AI Camera Capabilities

- Detects undefined objects in real-time to prevent hazards.
- Automatically counts people and vehicles for accurate occupancy and traffic data.
- Enhances situational awareness with continuous monitoring and alerts.
- Supports safety and operational efficiency through intelligent visual data analysis.

Privacy and Compliance

- Fully GDPR and KVKK compliant to ensure data privacy and security.
- Implements automatic rule violation controls using audit templates.
- Transforms images into actionable rules for safe and efficient process monitoring.
- Maintains user privacy while enabling comprehensive safety oversight.

Safety

Clothing, Equipment Inspection & Restricted Area Control

Navbea’s solutions transform images captured by existing IP-based cameras into actionable rules using customizable audit templates. This enables automatic monitoring and control of clothing and equipment compliance, ensuring workers adhere to safety standards in real-time. Restricted area control is enhanced through continuous surveillance, with immediate alerts for unauthorized access or rule violations. These automated systems streamline safety audits, reduce human error, and significantly improve overall process safety and compliance within facilities.

ESG

ESG Impact of RTLS

01 Environmental Monitoring

RTLS tracks real-time environmental data such as emissions and waste, enabling companies to optimize routes, reduce fuel consumption, and minimize their carbon footprint effectively.

02 Social Responsibility

The technology enhances workplace safety through AI-driven accident prevention and supports community engagement by providing data for social initiatives and improving emergency response.

03 Governance and Reporting

By offering verifiable data on operations, RTLS supports transparent ESG reporting, compliance tracking, and improved supply chain governance, boosting accountability and informed decision-making.

ESG

Environmental, Social, and Governance Solutions

01

Reducing Environmental Footprint

RTLS technology tracks emissions and optimizes logistics routes, reducing fuel consumption and carbon emissions, thereby supporting corporate sustainability goals.

02

Enhancing Community Engagement

RTLS enables real-time information sharing about local events and services, and supports community initiatives like environmental cleanups and social programs.

03

Compliance and Risk Management

Real-time monitoring aids in tracking adherence to environmental regulations and governance standards, helping organizations mitigate operational risks.

04

Transparent ESG Reporting

Accurate sensor data and analytics provide verifiable insights for ESG reporting, increasing accountability and stakeholder trust through transparent documentation.

Product

Sensor Variety and Product Applications



Diverse Sensor Types

Over 100 sensor types including active and passive sensors provide comprehensive data collection capabilities.



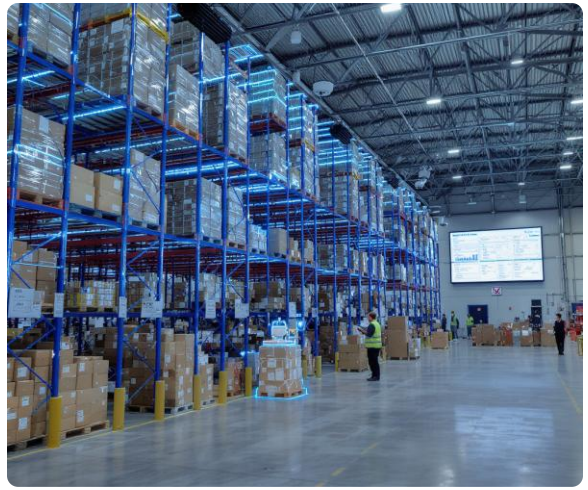
Product Monitoring

Sensors track product quality, location, and storage conditions to ensure optimal management and reduce losses.



Equipment and Asset Tracking

Specialized sensors monitor equipment usage, location, and maintenance needs to improve operational efficiency.



Warehouse and Facility Management

Sensor networks enable real-time tracking of inventory, environmental conditions, and workflow optimization in facilities.

Central Management Overview



Vehicle and Forklift Coordination

Real-time tracking and task management improve logistics, reduce idle time, and optimize fuel usage for vehicles and forklifts.



Production Facility Oversight

Centralized monitoring of production lines ensures operational efficiency, timely maintenance, and minimizes downtime.



Human Resources and Safety Management

Integrates personnel tracking, safety alerts, and access control to enhance workforce security and compliance.

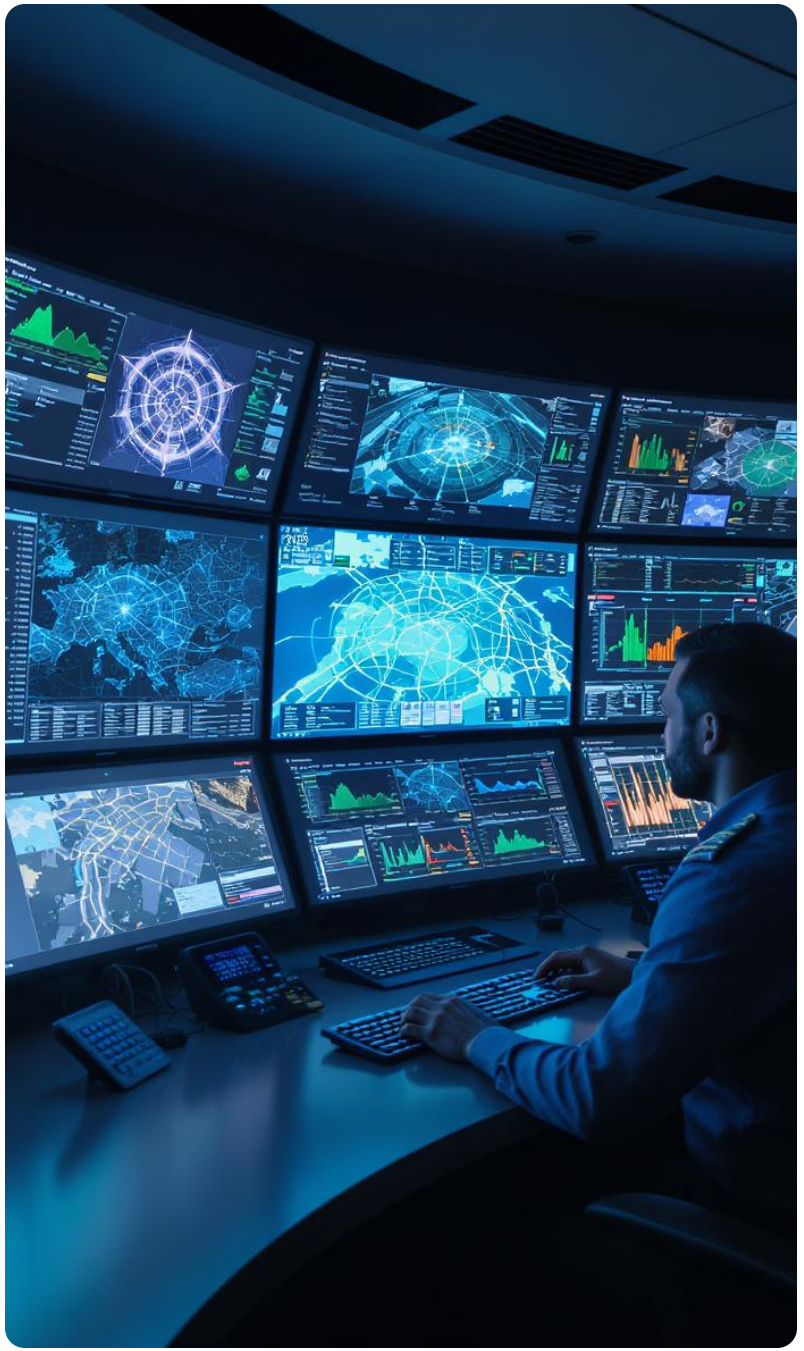


Productivity and Customer Insights

Combines productivity metrics and customer data for comprehensive performance analysis and improved service delivery.

Summary and Benefits

- Increases operational efficiency through real-time tracking and data-driven workforce and asset management.
- Enhances workplace safety with emergency alerts, AI-supported monitoring, and lone worker protection features.
- Reduces operational costs by minimizing equipment loss, optimizing maintenance, and improving resource allocation.
- Integrates advanced technologies such as IoT sensors, AI computer vision, and GPS for comprehensive monitoring.
- Supports sustainability and governance with data for environmental impact tracking and ESG reporting.



Thank You

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